

Contrasting GSM8K Accuracy Across Low-Bit Mixtral and Qwen3 GGUFs

Matthew Schwartz

Independent researcher [ORCID 0009-0009-4171-7247](https://orcid.org/0009-0009-4171-7247) matthew.schwartz.research@gmail.com

September 2026

Abstract

Choosing a smaller quantized model file saves storage and can change which layers fit on a GPU, but its accuracy must be measured. We evaluated Q4, IQ2, and IQ1 GGUF files for two mixture-of-experts models on the same 100 GSM8K arithmetic problems, using one llama.cpp build, fixed generation settings, and recorded CPU/GPU placements. Mixtral-8x7B scored 68%, 46%, and 26% at Q4, IQ2, and IQ1, whereas Qwen3-30B-A3B scored 91%, 97%, and 96%. Relative to Q4, the IQ2 and IQ1 files were 56% and 62% smaller for Mixtral and 44% and 48% smaller for Qwen. For Mixtral, Q4 was 22 and 42 percentage points higher; both paired differences remained significant after Holm correction across four comparisons. Qwen’s smaller-file point estimates exceeded Q4, but neither paired comparison was rejected after correction (both Holm $p = 0.0625$). This does not establish improvement or equivalence. The six-file benchmark documents substantially different observed size–accuracy tradeoffs. The lower tiers are not precision-matched between models, so the contrast cannot isolate architectural robustness. The experiment, P4R1, is corrective and not independent: it replaces a historical campaign with insufficient endpoint and model-identity provenance, whose outcomes and hypothesis were already known. Only the replacement measurements support the reported results.

1 Introduction

People running models locally often choose among several GGUF files for the same model. In the two sets studied here, the Q4 file uses more storage and memory than the IQ2 or IQ1 files, while a smaller file may fit more of the model in limited GPU memory. That trade is worthwhile only if the smaller deployment remains accurate enough for the intended task. File size is visible before download; the accuracy cost is not.

A quantization-tier label cannot supply that missing answer. Labels such as Q4, IQ2, and IQ1 name recipes, not a uniform precision for every tensor or a portable accuracy penalty. Prior work reports both strong low-bit MoE results and substantial variation across architectures and quantization methods [1–4]. The deployed result can also depend on the exact public file, inference engine, generation settings, and the engine’s placement of model layers between GPU and system memory.

We measure the size–accuracy tradeoff for Mixtral-8x7B and Qwen3-30B-A3B, using GSM8K grade-school arithmetic problems. Each of the six files answered the same 100 items through one llama.cpp build and a fixed generation and scoring protocol. Within each model, pairing correctness on the same questions shows both the aggregate difference and which deployment was correct when the other was not.

The answer differs between the two tested models. Mixtral scored 68% at Q4, 46% at IQ2, and 26% at IQ1; both Q4-to-low differences remained significant after correction for four comparisons.

Model	Artifact	File (GB)	Effective bpw	Experts/active	GPU layers/total
Mixtral-8x7B	Q4_K_M	28.4	4.87	8/2	12/33
	IQ2_XXS	12.6	2.15	8/2	26/33
	IQ1_M	10.8	1.86	8/2	29/33
Qwen3-30B-A3B	Q4_K_M	18.6	4.86	128/8	49/49
	UD-IQ2_XXS	10.3	2.71	128/8	49/49
	UD-IQ1_M	9.7	2.54	128/8	49/49

Table 1: The six P4R1 deployments, ordered from Q4 to the smaller IQ2 and IQ1 files. Decimal GB is calculated from exact artifact bytes; effective bpw uses the whole file and enumerated GGUF tensor shapes. The GPU column is llama.cpp’s offloaded-layer counter, which includes partially offloaded layers; 49/49 does not imply that all model weights reside in VRAM.

Qwen scored 91%, 97%, and 96%, respectively. Its smaller-file point estimates were higher than Q4, although neither comparison was rejected after Holm correction. One model’s result therefore does not tell a local user what the same tier choice will do for another model. The smaller files also use different effective precisions: Mixtral IQ1 and IQ2 have 1.86 and 2.15 whole-file bits per parameter, while Qwen’s corresponding files have 2.54 and 2.71. We compare the available files within each model; their labels do not define a matched cross-model intervention.

The contribution is an item-paired benchmark of six named deployments, including the exact files, observed placement, and reproducible statistical analysis. Section 2 explains the corrective study design and defines the deployments. Section 3 reports the paired results, and the remaining sections explain their practical meaning, limits, and reproducibility.

2 Methods

2.1 Study status and scope

P4R1 is a corrective exploratory experiment, not independent confirmation. A historical campaign did not preserve enough endpoint, model-identity, and placement provenance to attribute its saved scores reliably. We replaced all six GSM8K cells under a fixed protocol with explicit identity checks. The historical outcomes, sampled items, and motivating hypothesis were already known, which limits confirmatory interpretation even though the replacement protocol was frozen before execution. Only these six replacement cells contribute results. Historical MATH and HumanEval+ measurements are retained as audit context and are outside the present study.

2.2 Models and artifacts

The base checkpoints are Mixtral-8x7B-Instruct-v0.1 and Qwen3-30B-A3B-Instruct-2507, abbreviated below as Mixtral and Qwen. GGUF is the model container used by llama.cpp for CPU, GPU, and hybrid inference [5]. Labels such as IQ1_M, IQ2_XXS, and Q4_K_M name quantization recipes. They do not assert that every stored tensor uses one, two, or four bits. We therefore characterize each deployment by its exact file and by effective whole-file bits per stored parameter,

$$\text{effective file bpw} = \frac{8 \text{ (artifact bytes)}}{\text{parameters enumerated from GGUF tensor shapes}}. \quad (1)$$

This measure includes tensor-specific allocations and file overhead. It does not isolate routed-expert precision.

Mixtral uses repository revision 0e8d21e9cc0d28b59173d329348f60f8628d1c05; Qwen uses revision eea7b2be5805a5f151f8847ede8e5f9a9284bf77 [6, 7]. The campaign binds each basename, byte count, and SHA-256 digest to a frozen manifest and verifies content again after all six cells.

The artifacts within a ladder share a base checkpoint and publisher. The ladders do not share model lineage, quantization recipe, effective precision, or expert geometry.

2.3 Dataset, generation, and scoring

The evaluation uses 100 examples sampled without replacement from the GSM8K test split [8]. The source revision and subset digest are frozen. Sampling used Python’s deterministic random generator with seed 42. Every cell received the same item identifiers in the same evaluation contract.

Each request contained one user message: the GSM8K question followed by “End your response with: Answer: <number>”. There was no separate system message and no worked-example prompt. The serving engine applied each artifact’s chat template.

All six cells used a 4096-token context, a maximum of 2048 generated tokens, temperature 1.0, top- p 0.95, top- k 20, and concurrency 1. The per-item sampler seed was 4,200,000 plus the integer GSM8K item suffix. Thus each item used the same per-item seed shared across all six artifacts. The request specified `medium` reasoning effort. No cell dropped the chat-template keyword arguments.

The scorer extracts the final numeric answer and compares it with the gold value using absolute and relative tolerance 10^{-4} . Private outputs and gold answers were recomputed during promotion. A request failure, malformed response, parser failure, or missing answer was scored wrong rather than omitted. The promoted rows contain no request errors. The output limit was reached once in Mixtral IQ1 and once in each Qwen cell; these rows remain in their assigned cells.

2.4 Engine, placement, and lifecycle evidence

All cells ran through the same llama.cpp executable from clean commit 035e22731a7fd70b9854b3a2d64ec68e9b1a45d3, built for CUDA 12.6 and the target GPU architecture. The executable, build cache, compiler tools, CUDA libraries, and driver were hashed or read back before inference. The machine used an Intel Core i9-14900HX, 62 GiB of RAM, and an NVIDIA RTX 4080 Laptop GPU with 12282 MiB of VRAM at an 80 W power limit.

Cells ran sequentially with one server at a time. Before scoring, the guard verified the intended model hash and endpoint alias, rejected stale listeners or foreign compute processes, and captured the engine’s placement readback. Mixtral reported 12/33, 26/33, and 29/33 GPU-offloaded layers for Q4, IQ2, and IQ1. Qwen’s offloaded-layer counter was 49/49 for every tier, but it includes partially offloaded layers: automatic fitting reported system-memory overflow for 25 Q4 layers and two IQ2 layers. IQ1 required no fit adjustment. Thus the 18.6 GB Q4 file was not wholly GPU-resident, and the matching counters do not establish matched tensor placement. The artifact includes hash-bound startup excerpts and separate CPU-mapped and CUDA model-buffer readbacks; these buffer sizes are not additive measurements of physical resident memory.

Each cell has a startup receipt, completion receipt, and shutdown receipt; the next server could not begin until the previous process tree and listener were gone. Post-campaign verification rechecked model content and the CUDA runtime.

These controls address the specific failure of the historical run: P4R1 binds the response stream to the intended endpoint, exact model artifact, observed placement, and server lifecycle. They do not turn placement into a randomized factor. In particular, the Mixtral artifact tiers also differ in the number of layers placed on the GPU.

2.5 Statistical analysis

For each cell we report exact accuracy and a 95% Wilson interval [9]. Within a model, Q4 and each low tier answer the same items. Let $Y_{i,Q4}$ and $Y_{i,L}$ denote binary correctness for item i . The

Model	Artifact	Correct / 100	Accuracy [95% Wilson interval]
Mixtral-8x7B	Q4_K_M	68	68% [58.3, 76.3]
	IQ2_XXS	46	46% [36.6, 55.7]
	IQ1_M	26	26% [18.4, 35.4]
Qwen3-30B-A3B	Q4_K_M	91	91% [83.8, 95.2]
	UD-IQ2_XXS	97	97% [91.5, 99.0]
	UD-IQ1_M	96	96% [90.2, 98.4]

Table 2: GSM8K accuracy on the same 100 items in all six deployments. Wilson intervals describe individual cell proportions; paired comparisons appear in Table 3.

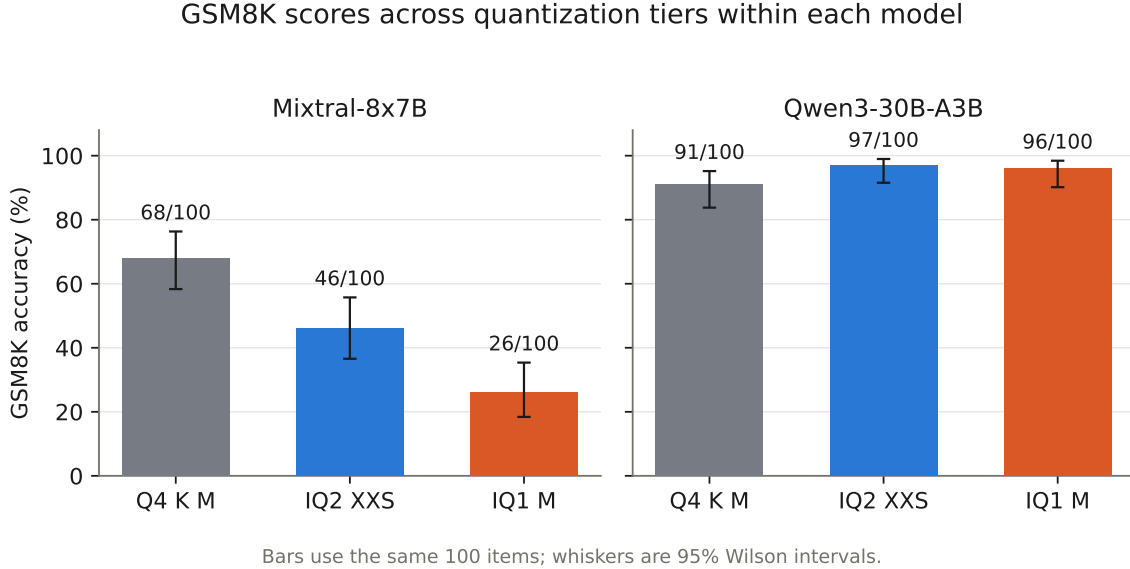


Figure 1: Within-model GSM8K scores. Bars share the same 100 items; whiskers are 95% Wilson intervals. Overlap or non-overlap of these marginal intervals is not the paired test.

paired estimand is

$$\hat{\Delta}_{Q4-L} = \frac{1}{n} \sum_{i=1}^n (Y_{i,Q4} - Y_{i,L}). \quad (2)$$

Positive values mean Q4 scored higher; negative values mean the low tier scored higher.

We report Newcombe method-10 paired 95% intervals for $\hat{\Delta}_{Q4-L}$ and exact two-sided McNemar tests [10, 11]. Holm’s step-down procedure controls family-wise error across exactly four comparisons [12]. The Newcombe intervals are pointwise; they are not adjusted simultaneous intervals. The family-wise decision follows the Holm-adjusted McNemar values. We did not predefine an equivalence margin, so failure to reject is not an equivalence result.

3 Results

3.1 The contrast across the six deployments

Table 2 and Figure 1 show the observed scores. The Mixtral results fall sharply as file size decreases; Qwen’s smaller files have higher point estimates. The paired comparisons below quantify those differences and their uncertainty. They test each smaller file against its within-model Q4 anchor, rather than testing an interaction between model family and quantization tier.

Comparison	Signed direction	Q4-low pp [95% CI]	Q4-only	Low-only	Holm p
Mixtral IQ2	Q4 scored higher	22 [10.4, 32.6]	30	8	0.00141596
Mixtral IQ1	Q4 scored higher	42 [30.4, 51.7]	45	3	5.25×10^{-10}
Qwen IQ2	the low tier scored higher	-6 [-12.3, -1.3]	0	6	0.0625
Qwen IQ1	the low tier scored higher	-5 [-10.9, -0.5]	0	5	0.0625

Table 3: Paired Q4-to-low comparisons. The signed difference is Q4 accuracy minus low-tier accuracy. Intervals are paired Newcombe method 10; Holm values adjust the four exact two-sided McNemar tests.

3.2 Mixtral: the smaller files scored lower

Choosing IQ2 saved 56% of the Q4 file size and reduced observed accuracy by 22 percentage points. Choosing IQ1 saved 62% and reduced it by 42 points. The paired evidence is asymmetric: Q4 answered 30 items that IQ2 missed, while IQ2 answered 8 that Q4 missed. The corresponding Q4-only and IQ1-only counts were 45 and 3. Thus the losses concern many of the same questions, rather than a comparison of unrelated samples.

Both exact McNemar tests reject after Holm adjustment at $\alpha = 0.05$. Table 3 reports the paired intervals and adjusted values; the unadjusted values are $p = 4.72 \times 10^{-4}$ for IQ2 and $p = 1.313 \times 10^{-10}$ for IQ1. The measured storage saving came with a substantial GSM8K accuracy cost for these Mixtral deployments.

3.3 Qwen: the low-tier point estimates exceed Q4

Qwen IQ2 and IQ1 saved 44% and 48% of the Q4 file size. Their scores exceeded Q4’s 91/100: 6 discordant items favored IQ2 and 5 favored IQ1, with none favoring Q4 in either comparison. In the Q4-minus-low convention, the differences are -6 and -5 points.

The unadjusted exact McNemar values are $p = 0.03125$ and $p = 0.0625$. Both become $p = 0.0625$ after Holm adjustment, so neither comparison is rejected within the four-test family. The pointwise Newcombe intervals in Table 3 are not simultaneous intervals or inversions of the exact tests; they can exclude zero while the adjusted tests do not reject. These data record the higher low-tier point estimates without establishing either improvement or equivalence.

4 Discussion

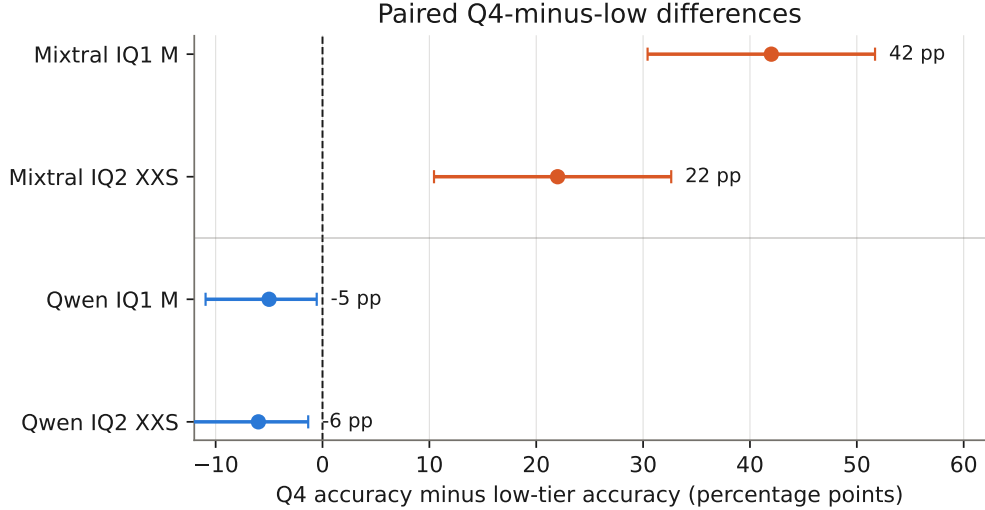
4.1 The practical answer

The file-size tradeoff did not have one accuracy answer across the two tested models. The Mixtral Q4 file answered substantially more of the same questions correctly than either smaller file. For Qwen, both smaller-file point estimates exceeded Q4, but the Holm procedure did not reject either comparison at a 0.05 family-wise threshold. A local user therefore cannot transfer the Mixtral result to Qwen, or the Qwen point estimates to another model, from tier names alone.

This is a deployment result, not a universal MoE quantization rule or a tested cross-model interaction. The paired tables add information that six marginal percentages would miss. Mixtral’s Q4-only counts greatly exceed its low-only counts. The Qwen discordances in this sample all run in the opposite observed direction. That is why the results should be reported as signed paired outcomes. At the same time, only five or six Qwen items were discordant, so multiplicity-aware inference remains limited.

4.2 Why the contrast does not identify a cause

The effective precisions differ before architecture enters the comparison. Qwen’s IQ1 file uses 2.54 whole-file bpw, more than even Mixtral’s IQ2 file at 2.15 bpw. The Q4 files are close in effective bpw,



Whiskers are paired Newcombe method 10 95% intervals; n=100 per comparison.

Figure 2: Paired Q4-minus-low differences with pointwise Newcombe method-10 95% intervals. Positive values favor Q4; negative values favor the lower tier. Family-wise decisions use the Holm-adjusted values in Table 3.

but matching that anchor does not match the amount or tensor-level distribution of compression at the lower tiers. This alone prevents reading the two ladders as a controlled test of low-bit robustness.

The architectures also differ in expert granularity: 8 experts with 2 active for Mixtral, versus 128 with 8 active for Qwen. Fine-grained MoE designs are intended to partition capacity and routing more flexibly [13, 14]. It is plausible that expert width, routing alternatives, or tensor allocation changes how weight error affects a model.

P4R1 compares one checkpoint from each family. Expert count changes together with training, model width, router design, active-expert count, quantization recipe, and placement. A follow-up could test granularity with multiple checkpoints and controlled precision and placement. The present observations cannot favor that explanation over the other differences.

4.3 Implications for local deployment

The smaller file can ease a storage or memory constraint and may allow more layers to be assigned to the GPU. The present Mixtral offload counters show that the engine assigned more layers to the GPU for IQ2 and IQ1 than for Q4. This paper measures the other half of the decision—whether the smaller deployment still answers the target task correctly enough. It does not combine speed and accuracy into a single recommendation.

The relevant unit is the exact artifact and engine under its actual memory placement. Effective file bpw is more informative than a tier label, but it still cannot predict accuracy by itself. A sound local evaluation should hash the artifact, verify the serving endpoint and loaded model, record placement, keep the task and generation protocol fixed, and pair responses on the same items. This is especially important when the GPU cannot hold every artifact under the same placement.

The result also cautions against treating Q4 as a universal quality ceiling. It is a practical within-model anchor. P4R1 includes no unquantized or near-lossless reference, so it cannot determine which artifact is closest to the underlying checkpoint’s maximum GSM8K accuracy.

5 Threats to Validity

P4R1 is not independent. The historical outcomes, exact item set, and hypothesis were known before the corrective run. The protocol and analysis were frozen for P4R1, but the resulting evidence is best interpreted as a provenance-corrected exploratory measurement. An independent confirmation would require newly sampled items, prospectively fixed hypotheses, and preferably additional model families.

The historical campaign’s endpoint, model-identity, and placement provenance was insufficient. P4R1 repairs those links for the replacement measurements; it cannot validate the earlier scores. Historical MATH and HumanEval+ results remain outside this study.

The evaluation covers one task, 100 items, one seeded decode per item and artifact, and one machine. It does not estimate run-to-run sampling variability. A few responses reached the output limit, although no promoted request was dropped and failures would have counted as wrong. GSM8K numeric-answer accuracy does not measure coding, factual recall, dialogue quality, calibration, or long-context behavior.

The comparisons are deployed configurations rather than isolated quantization treatments. Mixtral placement changes from 12/33 layers at Q4 to 26/33 and 29/33 at the lower tiers. Qwen’s matching 49/49 counters also conceal different system-memory overflow and model-buffer allocations. Model family, artifact recipe, and tensor allocation differ across the ladders. Effective whole-file bpw does not reveal expert-tensor precision. Consequently, neither cross-model differences nor within-ladder changes identify a quantization algorithm’s causal effect in isolation.

Finally, the paired intervals are pointwise while the hypothesis decisions use Holm-adjusted p values. No simultaneous confidence intervals, equivalence margin, power target, or preregistered minimum effect of interest was specified. The Qwen results therefore support their observed signed point estimates and the stated family-wise non-rejections, not interchangeability among artifacts.

6 Related Work

MoE architectures reduce active computation by routing tokens through a subset of experts [15, 16]. DeepSeekMoE and Qwen develop finer-grained expert organizations that motivate expert granularity as a variable of interest [13, 14]. Offloading work addresses the systems problem of serving MoEs when all weights do not reside in fast device memory [17].

MoQE uses expert-aware mixed precision and an optimized runtime [1]; QMoE co-designs a compression format and GPU decoding kernels [2]. QuantMoE-Bench studies structure-aware bit allocation, and Q-Strata optimizes hierarchical mixed-precision allocation across MoE blocks [4, 18]. These are quantization or execution methods. P4R1 introduces no corresponding optimizer, quantizer, or kernel.

Colla-Q allocates expert precision using activation entropy and evaluates Mixtral at average bit-widths from 1.57 to 2.54 with GPTQ-based quantization [19]. Its benchmark suite and bit allocation differ from the downloadable GGUFs tested here. Such method-level results reinforce why our Mixtral losses cannot be generalized to every low-bit representation of that model.

Evaluations spanning models and quantizers already report substantial variation, including low-precision fragility in Mixtral [3, 4]. A pre-existing public GGUF Quantization Benchmark also compares tiers within a model under llama.cpp and explicitly frames the practical question as the accuracy lost when moving to a smaller GGUF [20]. MoE-XBench evaluates the same Qwen3-30B-A3B-Instruct-2507 checkpoint in llama.cpp: its Q4_K_M GSM8K score is 0.833 against 0.836 in BF16. Its low-bit sensitivity analysis uses perplexity and excludes IQ1_M from the primary quantization-sensitivity aggregate; the paper also reports IQ1_M runtime measurements [21]. It does not report the item-paired GSM8K comparison for our six public files. Even where the

checkpoint and task overlap, its generated artifacts and evaluation protocol differ. The scores cannot be combined with ours or treated as a matched replication. We make no general priority claim for within-model GGUF benchmarking. The narrower contribution here is item-paired GSM8K evidence with artifact, engine, placement, and lifecycle provenance for these six exact deployments; it is not an expert-count scaling law.

7 Reproducibility and Artifact Availability

The version-1.0.0 reproducibility compendium is identified by [doi:10.5281/zenodo.22346482](https://doi.org/10.5281/zenodo.22346482); the corresponding source and report use the [versioned research-artifact directory](#). It contains 600 sanitized item-level outcomes, frozen claims, exact model and dataset identities, engine/build and placement metadata, the replication sources, and this report. Benchmark questions and answers, response prose, model weights, runtime binaries, and private lifecycle logs are excluded. The evidence records bind the public projection to the checked private campaign; the public archive does not expose the private logs themselves.

Analysis reproduction needs only Python’s standard library. From the artifact root, `python3 check_claims.py` recomputes the six cell counts, four discordance tables, Wilson and Newcombe intervals, exact McNemar values, and Holm adjustment. The paper rebuild uses the shipped figures and cached Tectonic 0.17.0. Full replication instead obtains the pinned upstream GSM8K revision and six model files, verifies their hashes, and runs one server at a time under the documented protocol. It requires compatible hardware, 84.23 GiB of model files, and local review of memory and thermal limits.

8 Conclusion

Choosing a smaller GGUF file did not produce a uniform accuracy cost across the two tested models. On the same 100 GSM8K items, Mixtral scored 68% at Q4, 46% at IQ2, and 26% at IQ1; both Q4-to-low paired differences remained significant after Holm correction. Qwen scored 91%, 97%, and 96%. The smaller-file point estimates were above Q4, but neither comparison was rejected after Holm correction. The Qwen non-rejections establish neither improvement nor equivalence.

The useful result is the measured storage–accuracy tradeoff for these exact files on this task. Lower-tier precision differs between the model families, and the experiment does not isolate an architectural cause. A broader recommendation requires newly sampled tasks and items, additional checkpoints, and matched precision and placement.

AI assistance. Claude (Anthropic) assisted with evaluation harness development and experiment orchestration. Claude and Codex (OpenAI) assisted with evidence auditing, analysis and figure code, citation checking, and manuscript drafting and revision. Matthew Schwartz directed the research and is responsible for its methods, results, interpretation, claims, citations, rights, and released artifacts.

References

- [1] Young Jin Kim, Raffy Fahim, and Hany Hassan Awadalla. Mixture of quantized experts (MoQE): Complementary effect of low-bit quantization and robustness, 2023. URL <https://arxiv.org/abs/2310.02410>.
- [2] Elias Frantar and Dan Alistarh. QMoE: Practical sub-1-bit compression of trillion-parameter models, 2023. URL <https://arxiv.org/abs/2310.16795>.
- [3] Shiyao Li, Xuefei Ning, Luning Wang, Tengxuan Liu, Xiangsheng Shi, Shengen Yan, Guohao Dai, Huazhong Yang, and Yu Wang. Evaluating quantized large language models, 2024. URL <https://arxiv.org/abs/2402.18158>. Version 2.
- [4] Pingzhi Li, Xiaolong Jin, Zhen Tan, Yu Cheng, and Tianlong Chen. QuantMoE-Bench: Examining post-training quantization for mixture-of-experts, 2024. URL <https://arxiv.org/abs/2406.08155>.
- [5] Georgi Gerganov and llama.cpp contributors. llama.cpp: LLM inference in C/C++. Software repository, 2023. URL <https://github.com/ggml-org/llama.cpp>. Accessed 2026-09-05.
- [6] mradermacher. Mixtral-8x7B-Instruct-v0.1 weighted/importance-matrix GGUF quantizations. Hugging Face model repository, 2025. URL <https://huggingface.co/mradermacher/Mixtral-8x7B-Instruct-v0.1-i1->

- GGUF/tree/0e8d21e9cc0d28b59173d329348f60f8628d1c05. Repository revision 0e8d21e9cc0d28b59173d329348f60f8628d1c05; accessed 2026-09-05.
- [7] Unsloth. Qwen3-30B-A3B-Instruct-2507 dynamic GGUF quantizations. Hugging Face model repository, 2025. URL <https://huggingface.co/unsloth/Qwen3-30B-A3B-Instruct-2507-GGUF/tree/eea7b2be5805a5f151f8847ede8e5f9a9284bf77>. Repository revision eea7b2be5805a5f151f8847ede8e5f9a9284bf77; accessed 2026-09-05.
 - [8] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, 2021. URL <https://arxiv.org/abs/2110.14168>.
 - [9] Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. doi: 10.1080/01621459.1927.10502953. URL <https://doi.org/10.1080/01621459.1927.10502953>.
 - [10] Robert G. Newcombe. Improved confidence intervals for the difference between binomial proportions based on paired data. *Statistics in Medicine*, 17(22):2635–2650, 1998. doi: 10.1002/(SICI)1097-0258(19981130)17:22<2635::AID-SIM954>3.0.CO;2-C. URL [https://doi.org/10.1002/\(SICI\)1097-0258\(19981130\)17:22<2635::AID-SIM954>3.0.CO;2-C](https://doi.org/10.1002/(SICI)1097-0258(19981130)17:22<2635::AID-SIM954>3.0.CO;2-C).
 - [11] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947. doi: 10.1007/BF02295996. URL <https://doi.org/10.1007/BF02295996>.
 - [12] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. URL <https://www.jstor.org/stable/4615733>.
 - [13] Damai Dai, Chengqi Deng, Chenggang Zhao, R. X. Xu, Huazuo Gao, Deli Chen, Jiashi Li, Wangding Zeng, Xingkai Yu, Y. Wu, Zhenda Xie, Y. K. Li, Panpan Huang, Fuli Luo, Chong Ruan, Zhifang Sui, and Wenfeng Liang. DeepSeekMoE: Towards ultimate expert specialization in mixture-of-experts language models, 2024. URL <https://arxiv.org/abs/2401.06066>.
 - [14] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025. URL <https://arxiv.org/abs/2505.09388>.
 - [15] William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity, 2021. URL <https://arxiv.org/abs/2101.03961>.
 - [16] Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, Gianna Lengyel, Guillaume Bour, Guillaume Lample, L  lio Renard Lavaud, Lucile Saulnier, Marie-Anne Lachaux, Pierre Stock, Sandeep Subramanian, Sophia Yang, Szymon Antoniak, Teven Le Scao, Th  ophile Gerv  t, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. Mixtral of experts, 2024. URL <https://arxiv.org/abs/2401.04088>.
 - [17] Artyom Eliseev and Denis Mazur. Fast inference of mixture-of-experts language models with offloading, 2023. URL <https://arxiv.org/abs/2312.17238>.
 - [18] Deokjae Lee, Sihun Chu, and Hyun Oh Song. Q-Strata: Hierarchical bit allocation for mixed-precision quantization of mixture-of-experts LLMs, 2026. URL <https://arxiv.org/abs/2608.30564>. EMNLP 2026 (main conference, long paper), per the arXiv comment.
 - [19] Eunju Shin and Jongbin Ryu. Colla-Q: Toward collaborative experts in MoE quantization via minimax precision balancing, 2026. URL <https://arxiv.org/abs/2609.18131v1>. Submitted 2026-09-16; EMNLP 2026 acceptance reported in the arXiv comment; checked 2026-09-28.
 - [20] Yves Rougy. GGUF quantization benchmark. Software and benchmark repository, 2026. URL <https://github.com/yrougy/llm-quant-bench/tree/4d93dbc5c115019a09dca0ab6cb532b78bc9da81>. Repository snapshot dated 2026-08-25; accessed 2026-09-05.
 - [21] Afsara Benazir, Chen Chen, Rongxiao Qu, Jiabo Huang, Jingtao Li, and Lingjuan Lyu. Benchmarking composable compression techniques in mixture-of-experts LLMs, 2026. URL <https://arxiv.org/abs/2608.21693v1>. Version 1, submitted 2026-08-22; checked 2026-09-28.